

Harshavardhan Shet

Generative AI / ML Engineer

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Summary

Generative AI / Machine Learning Engineer with 2+ years of experience building AI solutions for automotive, edge, and cloud platforms. Skilled in on-device LLMs, LLM fine-tuning, Agentic AI, RAG, computer vision, and end-to-end model optimization across NVIDIA Jetson/DGX, Qualcomm Snapdragon, and Google Cloud (Vertex AI, TPU).

Technical Skills

Languages: Python, C++, Java, SQL

Frameworks: PyTorch, TensorFlow, HuggingFace Transformers, PEFT/LoRA, llama.cpp, LangChain, LangGraph, TensorRT-LLM, OpenCV, Pandas, NumPy, CUDA

Generative AI: LLM fine-tuning (LoRA), Agentic AI, Retrieval-Augmented Generation (RAG), prompt engineering, function calling, PageIndexing, vector databases (FAISS, LlamaIndex), model quantization (INT4/INT8)

Vision & Speech: object detection, semantic segmentation, speech-to-text (ASR), text-to-speech, speaker diarization

Edge & Hardware: NVIDIA Jetson Orin, NVIDIA DRIVE AGX, NVIDIA DGX (A100), Qualcomm Snapdragon 8 Gen 3, Google Cloud TPU

Cloud & MLOps: GCP (Vertex AI, Cloud TPU), Docker, Git, Linux, Flask, ONNX, TensorRT, SNPE, QNN, Genie SDK, DeepStream

Professional Experience

Generative AI Engineer

June 2024 – Present

Tata Consultancy Services — Automotive Gen AI Center of Excellence (R&D)

Bengaluru, India

GPU-to-TPU Pipeline Migration — Autonomous-Vehicle OEM

- Migrated a multi-model camera KPI tool (vision + LLM summarization) from NVIDIA DGX GPUs to GCP TPU VMs; wrote RF-DETR/YOLO conversion and quantization scripts, achieving 130 inferences/sec on multi-pod TPUs.
- Integrated Vertex AI-hosted LLMs to auto-summarize mAP-based KPI reports; Dockerized the application for customer deployment.

In-Vehicle Virtual Assistant — NVIDIA Jetson Orin

- Built a fully offline voice assistant for natural-language control of in-car functions (AC, fan speed, sunroof); deployed an on-board LLM via llama.cpp with local speech-to-text (STT) and text-to-speech (TTS) across shared GPU/CPU.
- Fine-tuned the on-board LLM with LoRA (PEFT) on curated in-vehicle command-function pairs, improving intent-classification and function-call accuracy over the base model; merged and quantized the adapter for on-device inference.
- Integrated agent-based workflows for real-time query answering and dynamic vehicle-settings control, enabling free-form commands instead of fixed phrases.

Personalized Driving Experience System — Premium Automotive OEM, Infotainment

- Architected a hybrid neural network + reinforcement learning system that learns driver routines from live drive data and auto-suggests seat, climate, route, and media settings at ignition; achieved ~85% prediction accuracy. **(Best Team Award)**
- Generated realistic synthetic driving-routine data with OpenStreetMap + LLMs, eliminating weeks of manual collection; quantized and compiled all models for edge — 37 ms/inference on GPU, all data on-device.

On-Device LLM Deployment — Qualcomm Snapdragon 8 Gen 3

- Deployed Llama 3.2 8B fully on-device using Genie SDK and QNN, owning conversion, INT8 quantization, compilation, profiling; achieved 32 tokens/sec with first-token latency under 200 ms; presented live at the Qualcomm AI Summit. **(On-the-Spot Team Award)**

Camera Lens Obstruction Segmentation — Warehouse Monitoring

- Fine-tuned a DINOv2-backbone segmentation model (ResNet encoder-decoder head) to detect water droplets, dust, and glare for camera-blockage pre-alerts; achieved 94% precision (+11 pts over baseline) via targeted augmentation; quantized for edge with dynamic batching.

LLM-Powered Test Case Generation Tool — Hardware OEM

- Built a requirements-to-test-cases tool with a 3-stage agentic pipeline (rules vector DB, retrieval agent, generation agent) on NVIDIA Nemotron hosted on DGX A100s; reduced manual effort by 60% and adopted by multiple customers as a top team offering.

PageIndex vs. RAG — Retrieval Research

- Benchmarked PageIndex (graph-based indexing) vs. RAG on retrieval accuracy, chunking, and cross-chunk context; pioneered swapping the ChatGPT API for a local llama.cpp-served LLM via a custom wrapper — no known prior public implementation.

Projects

Phishing Domain Detection — Smart India Hackathon National Finalist, New Delhi

December 2023

- Curated a balanced dataset of ~50,000 domains via web scraping; designed and trained a custom ANN achieving 88% binary classification accuracy.

Publication

Speech Denoising and Transcription to Improve Audio Communication using Deep Learning

April 2024

IJIRCE, Vol. 12, Issue 4 | DOI: 10.15680/IJIRCE.2024.1204224

- Co-authored a CNN-based pipeline for speech denoising (STFT), speaker diarization, and ASR transcription of call-centre audio; evaluated via SNR, PESQ, and WER.

Education

Bachelor of Engineering in Computer Science

2020 – 2024

Bapuji Institute of Engineering and Technology

CGPA: 8.0/10

Certifications & Awards

Certifications: IBM Data Science Professional Certificate (IBM); Machine Learning Specialization (Stanford & DeepLearning.AI); Programming, Data Structures & Algorithms in Python (NPTEL)

Awards: Team of the Month, Jun 2026 (NVIDIA CXO in-lab demo); Best Team Award (Personalized Driving Experience); On-the-Spot Team Award (On-Device LLM); Best Outgoing Student (2020–24); State-Level Coding Competition Winner